

IN4MATX 285:

Interactive Technology Studio

Programming: Coding with Files

Today's goals

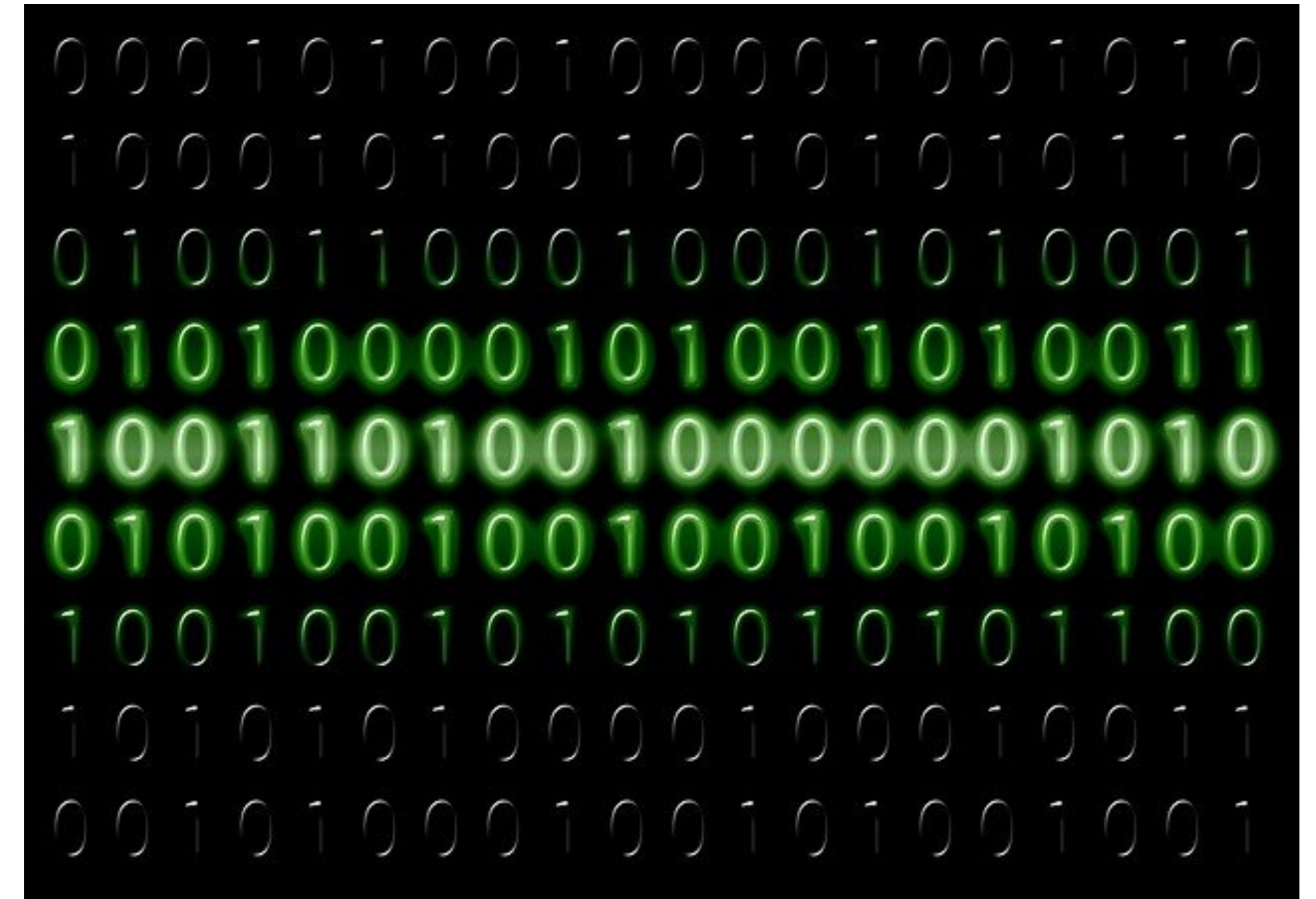
By the end of today, you should be able to...

- Use a code editor to edit a code file
- Create our first webpage and view it in a browser

What does code look like?

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- Maybe you've been told that code is the "binary" 0's and 1's that run on our computers
- While this is technically true, it's not how most people spend their time writing code
 - Way too burdensome; more on this near the end of the quarter



What does code look like?

- Instead, most code looks like text
 - A text document without any formatting (text size, spacing, etc.)
 - Maybe you've seen a .txt file
- What separates code from plain text is its *syntax*
 - E.g., a set of rules that tell your computer exactly what to do with the code



What does code look like?

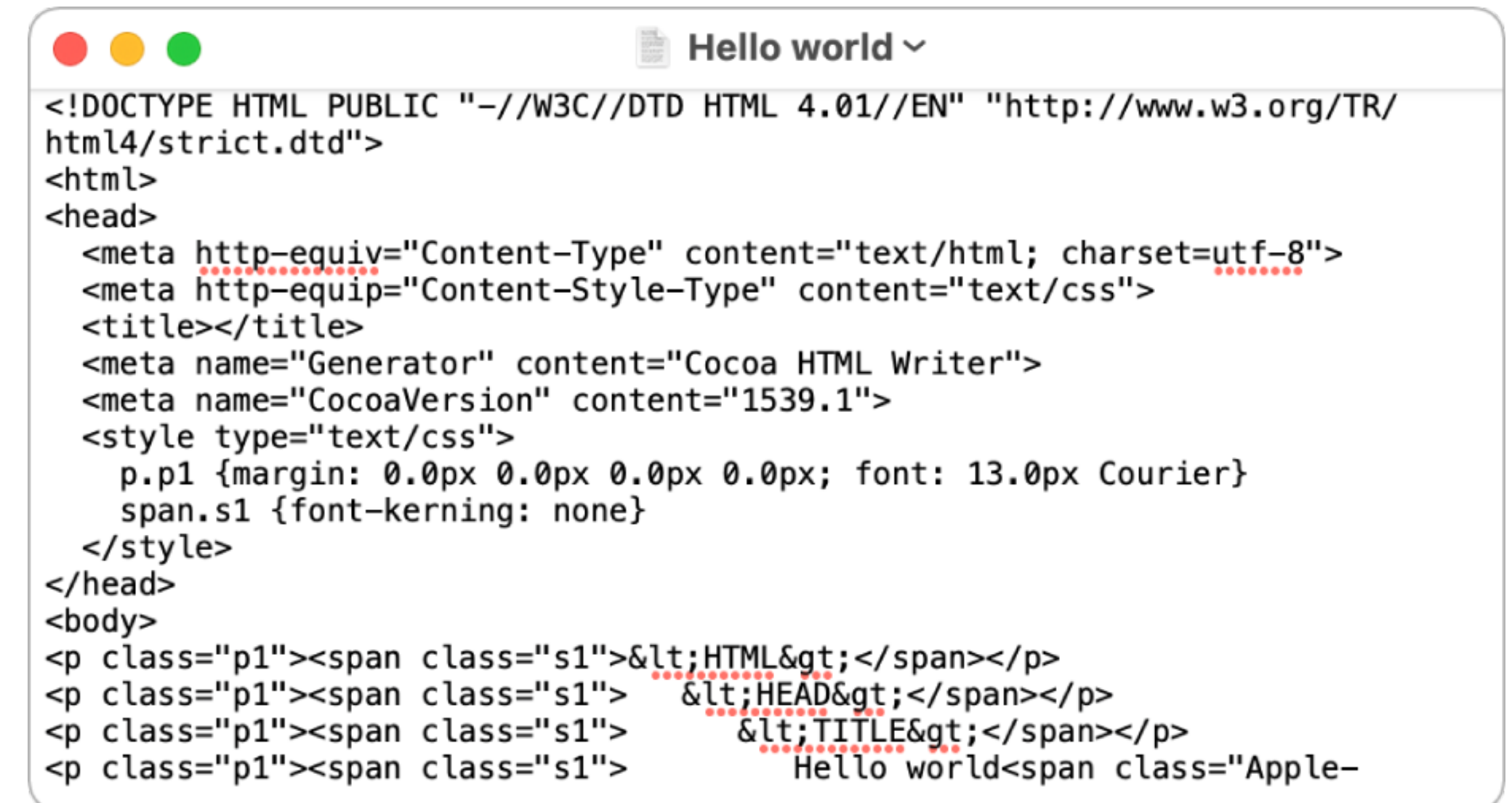
- File extensions
 - .txt, .docx, .pptx, .png, that sort of thing
 - These are used to signal what program should open these files, and what they should do with it
- Most code uses file extensions which correspond to the language
 - .html for HTML files, .css for CSS, etc.
 - But, you can open up any file in a text editor



How does one *edit* code?

Code editors

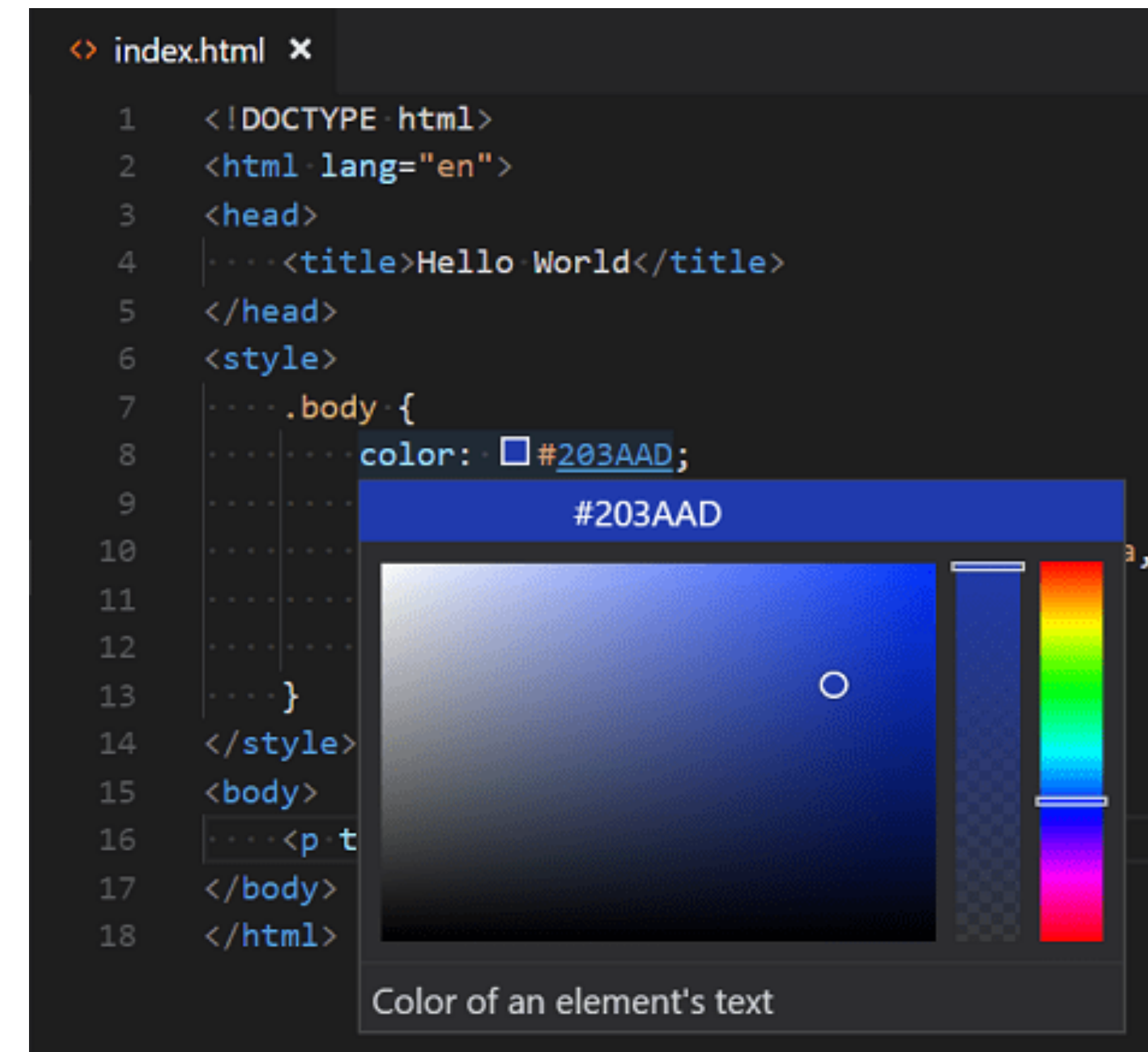
- Since code is just text, you can use any text editor to edit code
 - TextEdit on Mac, Notepad on Windows
- But, doing so gets annoying very quickly
 - Lack of formatting
 - Lots of repetitive typing (open tags, close tags)



```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
  <meta http-equiv="Content-Style-Type" content="text/css">
  <title></title>
  <meta name="Generator" content="Cocoa HTML Writer">
  <meta name="CocoaVersion" content="1539.1">
  <style type="text/css">
    p.p1 {margin: 0.0px 0.0px 0.0px 0.0px; font: 13.0px Courier}
    span.s1 {font-kerning: none}
  </style>
</head>
<body>
<p class="p1"><span class="s1">&lt;HTML&gt;</span></p>
<p class="p1"><span class="s1">    &lt;HEAD&gt;</span></p>
<p class="p1"><span class="s1">        &lt;TITLE&gt;</span></p>
<p class="p1"><span class="s1">            Hello world<span class="Apple-
```

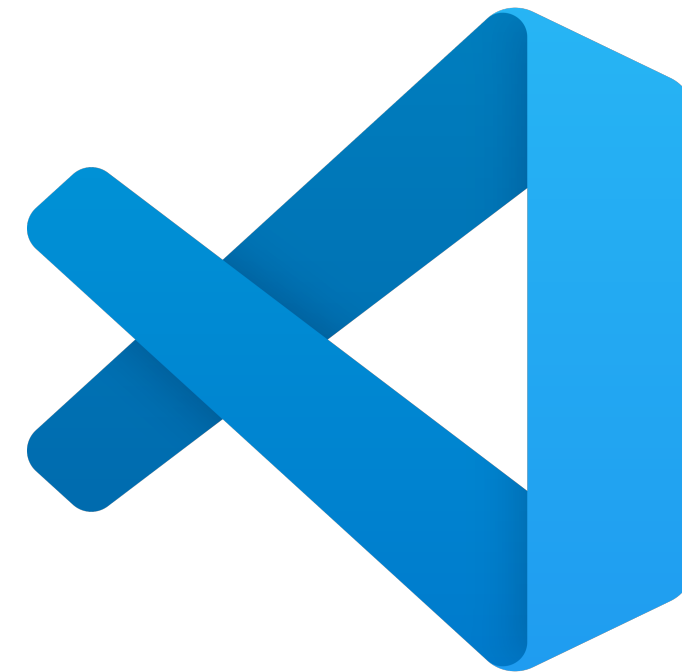

Code editors

- Instead, programmers use an Integrated Development Environment (IDE)
- IDEs provide:
 - Tools for checking syntax
 - Formatting assistance
 - Automatic text completion
 - Tools for compiling and debugging, sometimes



Code editors

- Widely-used IDEs
 - Visual Studio Code (Microsoft)
 - WebStorm (JetBrains)
 - Eclipse (Eclipse Foundation)
- If you're new to Web Development, we recommend using VS Code



Let's get coding!

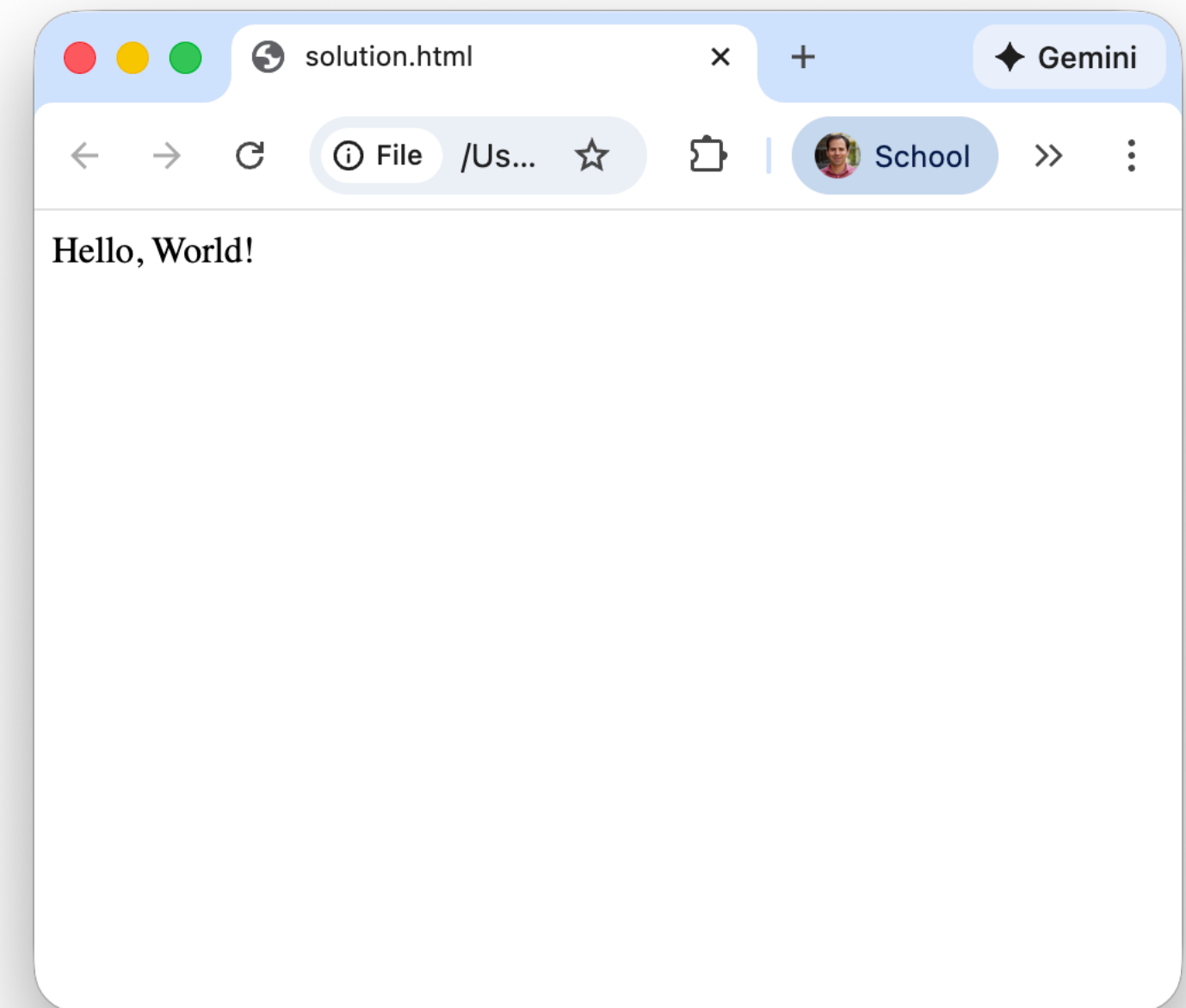
Live demos

- I'll post the starter and solution code on the class website
- I will try to make some mistakes to illustrate common pitfalls that people experience
- Feel free to ask follow-up questions about the demo



Hello, World!

- Typically, this is the first program that a person writes when they learn a new programming language
- We'll write a "Hello, World!" in HTML.



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